

THE TUNNEL EFFECT CORRECTION FOR PARABOLIC POTENTIAL BARRIERS *

BY R. P. BELL †

Chemistry Dept., Brookhaven National Laboratory, Upton, L.I., N.Y.

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An expression is derived for the permeability of a parabolic energy barrier, leading to expressions for the tunnel-effect corrections in chemical kinetics which should be valid even when these corrections are large.

The transition-state formulation of chemical kinetics involves a quantum correction which can be expressed in terms of the curvatures of the energy surface, and which is of particular importance for the theory of kinetic isotope effects. The energy surface has positive curvatures in most dimensions, corresponding to vibrations of the transition state, and the corresponding terms in the quantum correction involve the zero-point energies and the higher energy levels of these vibrations. These terms can be evaluated exactly in the harmonic approximation, and this has been done particularly by Bigeleisen, who has also given convenient expansions for computation.¹ In one dimension (the reaction co-ordinate) the energy surface has a negative curvature, corresponding to a maximum in the potential energy, and the quantum correction for this co-ordinate is the so-called tunnel effect. Much less is known about the tunnel-effect correction than about the zero-point energy terms, and the object of the present paper is to give a more accurate treatment for a parabolic energy-barrier.

The usual method of applying the tunnel correction rests on a general treatment due to Wigner.² He shows that for the passage of a particle of mass m over a col in an energy surface of n dimensions the first approximation to the quantum correction (assumed small) is

$$Q = v(\text{quantum})/v(\text{classical}) = 1 - \frac{\hbar^2}{4\pi^2} \sum_n \frac{A_i}{24mk^2T^2} \quad (1)$$

$$= 1 - \frac{1}{24} \sum_n \left(\frac{\hbar \nu_i}{kT} \right)^2,$$

where the A_i are the curvatures of the energy surface and the ν_i the corresponding frequencies. Vibrations of the transition state give positive values of A_i and real frequencies, and for these terms eqn. (1) is identical with the first term of the expression given by Bigeleisen. For the tunnel correction A_i is negative and ν_i imaginary, so that this part of the correction can be written as

$$Q = 1 + \frac{\hbar^2 A}{96\pi^2 mk^2 T^2} = 1 + (\hbar \nu/kT)^2/24, \quad (2)$$

where A is now positive, and the imaginary frequency is written as $i\nu$. This expression is satisfactory when the correction is small, but this will not always be the case in practice.

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† Visiting Senior Chemist from Balliol College, Oxford University, England.

An alternative treatment for a parabolic barrier has been given by Bell,³ whose results will be referred to as (I). In this treatment the permeability G of the barrier is obtained as a function of the energy W by using the B.W.K. approximation, leading to the expression

$$G = \exp \left\{ -\beta \left(1 - \frac{W}{E} \right) \right\}, \quad (W < E), \quad (3)$$

$$G = 1, \quad (W > E),$$

where

$$\beta = 2\pi^2 a (2mE)^{1/2} / \hbar$$

and E and $2a$ are the height and width of the barrier. The quantum correction to the reaction velocity is then obtained by integrating over an energy distribution, i.e.,

$$Q = \exp(E/kT) \int_0^\infty \frac{1}{kT} \exp(-W/kT) G(W) dW. \quad (4)$$

The B.W.K. approximation is valid only when W is considerably less than E , so that this treatment will be satisfactory when the degree of tunnelling is large. For a small degree of tunnelling energies in the neighbourhood of $W = E$ will be important and (3) will be a bad approximation: in particular, the assumption $G = 1$ for $W > E$ neglects the reflection of particles which takes place in this neighbourhood. In consequence the resulting expression for Q does not reduce to (2) when expanded in powers of $\hbar/kT$, but instead assumes the form $1 + \hbar\nu/2\pi kT + \dots$

Although no exact solution of the Schrödinger equation has been obtained for a parabolic barrier, there are a number of conditions which an exact expression for $G(W)$ must satisfy:

(i) It must reduce to the B.W.K. result (3) for $W \ll E$, and must tend to unity as $W \rightarrow \infty$.

(ii) An exact solution for the particular case $W = E$ gives $G(W) = 1/2$ to a good approximation for masses and barrier dimensions of chemical interest.⁴ This is confirmed when calculations are made⁵ by using the exact solution given by Eckart⁶ for a qualitatively similar type of barrier, and these calculations also suggest a symmetrical form for $G(W)$ such that $G(E - W) + G(E + W) = 1$.

These conditions are satisfied by the expression

$$G(W) = [1 + \exp(\beta y)]^{-1}, \quad y = 1 - W/E, \quad (5)$$

which must be close to the exact solution for problems of chemical interest. Substituting in (4), and writing $E/kT = \alpha$, we have

$$Q = \int_{-\infty}^1 \frac{\alpha \exp(\alpha y) dy}{1 + \exp(\beta y)}. \quad (6)$$

In problems of chemical kinetics α and β are both in the range 10-40, so that $e^{-\alpha}$ and $e^{-\beta}$ are both small, and $e^{\alpha-\beta}$ is usually either $\gg 1$ or $\ll 1$. There are then three cases:

(a) $\alpha > \beta$. Eqn. (6) can be integrated by putting $x = \exp\{(\alpha - \beta)y\}$, giving

$$Q = \frac{\alpha \exp(\alpha - \beta)}{\alpha - \beta} \{1 + O[\exp(-\beta)]\}, \quad (7)$$

the first term of which is identical with (I), as would be expected for a large degree of tunnelling. When Q is multiplied by the classical factor $e^{-\alpha}$ the reaction velocity is seen to vary only slowly with temperature: this case has not yet been encountered in chemical kinetics.

(b) $\alpha \approx \beta$. Substituting $x = \exp(\alpha y)$, (b) becomes

$$Q = \int_0^{e^\alpha} \frac{dx}{1 + x^{\beta/\alpha}} \quad (8)$$

If $\alpha = \beta$ this integrates to give

$$Q = \ln(1 + e^\beta) = \beta\{1 + O(e^{-\beta})\}, (\alpha = \beta), \quad (9)$$

which differs slightly from the result of (I), $Q = \beta + 1$. If α and β are not quite equal, (8) can be evaluated as a Taylor expansion in terms of $\partial Q/\partial\alpha$, $\partial^2 Q/\partial\alpha^2$, etc., about the point $\alpha = \beta$, giving

$$Q = \beta\{1 + \frac{1}{2}(\alpha - \beta) - \frac{1}{6}\beta(\alpha - \beta)^2 + \dots\} + O(e^{-\beta}), (\alpha \simeq \beta). \quad (10)$$

The same procedure can be used to obtain the temperature coefficient of Q when $\alpha = \beta$, and since the velocity constant can be written as $k = Ae^{-Q}$, this gives for the apparent activation energy E^* ,

$$E^*/E = 1 - d\ln Q/d\alpha = (\beta - 2)/2\beta, (\alpha = \beta), \quad (11)$$

again differing only slightly from (I), $E^*/E = \beta/2(\beta + 1)$.

(c) $\alpha < \beta$. This is the case which is most frequently of interest in chemical problems. Eqn. (8) can be written in the form

$$\begin{aligned} Q &= \int_0^\infty \frac{dx}{1 + x^{\beta/\alpha}} - \int_{e^\alpha}^\infty \frac{dx}{1 + x^{\beta/\alpha}} \\ &= \frac{\pi\alpha/\beta}{\sin \pi\alpha/\beta} - \frac{\alpha \exp(\alpha - \beta)}{\beta - \alpha} \left\{ 1 - \frac{\beta - \alpha}{2\beta - \alpha} \exp(-\beta) + \right. \\ &\quad \left. + \frac{\beta - \alpha}{3\beta - \alpha} \exp(-2\beta) - \dots \right\}. \end{aligned} \quad (12)$$

If $\exp(\alpha - \beta) \ll 1$ the second half of (12) can be neglected, giving

$$Q = \frac{1}{2} u / \sin \frac{1}{2} u \quad (u = 2\pi\alpha/\beta = h\nu/kT; u < 2\pi). \quad (13)$$

There are several points of interest in this result, which differs radically from (I) ($Q = \beta/(\beta - \alpha)$). Expansion in powers of u gives

$$Q = 1 + \frac{u^2}{24} + \frac{7u^4}{5760} + \dots, (u < 2\pi), \quad (14)$$

$$\ln Q = \frac{u^2}{24} + \frac{u^4}{2880} + \dots, (u < 2\pi). \quad (15)$$

The first term of the expansion is identical with Wigner's result (2), but validity of eqn. (13)-(15) should extend to much higher values of u . Moreover, the present result extends in a remarkable way Wigner's analogy between the two types of quantum correction, since (13) is identical with the exact expression for the harmonic oscillator contribution, $\frac{1}{2} u / \sinh \frac{1}{2} u$, if the real frequency of the oscillator is replaced by the imaginary one corresponding to the parabolic barrier. Similarly, the expansion (15) is identical with that given by Bigeleisen for an oscillator, except that the terms are now all positive instead of being alternately positive and negative.

Eqn. (13) leads directly to expressions for the apparent activation energy E^* and the apparent pre-exponential factor A^* , which are defined in terms of the observed velocity constant k by the equations

$$E^* = kT^2 d \ln k / dT, \quad \ln A^* = \ln k + E^*/kT. \quad (16)$$

The results are

$$E^* - E = kT(\frac{1}{2} u \cot \frac{1}{2} u - 1), \quad (17)$$

$$\frac{A^*}{A} = \frac{\frac{1}{2} u}{\sin \frac{1}{2} u} \exp(\frac{1}{2} u \cot \frac{1}{2} u - 1), \quad (18)$$

again closely analogous to those for an oscillator.

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- ¹ Bigeleisen, *J. Chem. Physics*, 1949, **17**, 675. Bigeleisen and Wolfsberg, *ibid.*, 1953, **21**, 1972. Bigeleisen *Proc. Int. Symposium on Isotope Separation*, (N. Holland Publishing Co., 1958), p. 148.
- ² Wigner, *Z. physik Chem. B*, 1932, **19**, 203.
- ³ Bell, *Proc. Roy. Soc. A*, 1935, **148**, 241.
- ⁴ Bell, *Proc. Roy. Soc. A*, 1937, **158**, 128.
- ⁵ Bell, *Proc. Roy. Soc. A*, 1933, **139**, 466.
- ⁶ Eckart, *Physic. Rev.*, 1930, **35**, 1303.